

NATAL, Brazil

WMO: 825990

Lat: 5.911S

Lon: 35.248W

Elev: 169

StdP: 14.61

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	69.5	69.9	64.1	90.0	79.9	64.7	92.2	79.3	24.6	82.8	23.1	82.8	6.5	180	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	12.1	90.9	77.8	89.8	77.3	89.2	77.2	80.0	85.9	79.4	85.4	79.0	85.2	15.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	150.3	83.2	77.4	143.6	82.1	77.2	142.4	81.9	43.8	86.6	43.1	85.8	42.5	85.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	21.2	19.5	DB	67.6	92.1	1.6	2.7	66.5	94.0	65.5	95.7	64.6	97.2	63.5	99.2	(4)
(5)			WB	65.5	81.8	1.7	2.0	64.2	83.3	63.2	84.5	62.2	85.6	60.9	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	81.7	81.9	82.0	81.4	80.4	78.6	77.7	77.7	78.6	80.2	81.0	81.8	(6)	
	DBStd	2.36	1.76	1.64	1.75	1.75	1.73	1.93	1.72	1.69	1.81	1.78	1.61	1.57	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11043	984	894	993	941	943	859	857	859	859	938	931	985	(10)	
	CDD65	5568	519	474	528	491	478	409	392	394	409	473	481	520	(11)	
	CDH74	51744	5506	5104	5653	4880	4294	2980	2575	2768	3292	4425	4799	5468	(12)	
(13)	CDH80	16157	1822	1756	1988	1608	1340	795	618	715	918	1324	1482	1791	(13)	
(14)	Wind	WSAvg	10.1	10.1	9.6	8.7	8.2	8.5	9.1	10.2	11.2	11.7	12.0	11.4	11.0	(14)
(15) Precipitation	PrecAvg	51.4	2.1	4.2	7.2	7.9	7.1	7.7	6.9	3.4	2.1	0.9	0.8	1.1	(15)	
	PrecMax	83.3	7.4	10.7	15.8	18.9	13.8	16.8	16.9	9.5	6.5	5.4	4.6	4.3	(16)	
	PrecMin	29.2	0.0	0.1	1.1	2.2	2.3	2.2	1.6	0.5	0.1	0.0	0.0	0.0	(17)	
	PrecStd	15.2	2.0	3.3	3.9	4.5	3.3	4.4	3.3	2.4	1.5	1.1	1.1	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	91.4	91.6	91.4	90.0	88.7	88.0	88.1	89.6	89.9	90.7	91.3	(19)	
		MCWB	77.8	78.4	78.6	78.6	78.1	77.2	75.9	75.3	75.8	75.6	76.5	77.6	(20)	
	2%	DB	89.6	89.7	89.9	89.7	88.9	87.4	86.2	86.3	87.6	88.1	89.2	89.6	(21)	
		MCWB	77.4	77.5	77.8	78.1	77.7	76.6	75.1	74.4	75.1	75.5	76.2	76.9	(22)	
	5%	DB	88.1	88.2	89.0	88.2	87.7	86.0	84.6	85.2	86.0	86.8	87.6	88.0	(23)	
		MCWB	76.8	77.1	77.5	77.7	77.3	75.9	74.4	73.9	74.4	74.9	75.8	76.4	(24)	
	10%	DB	87.2	87.7	87.8	87.4	86.2	84.4	83.9	84.1	84.5	85.9	86.2	86.4	(25)	
		MCWB	76.5	76.9	77.3	77.6	77.1	75.4	74.2	73.5	73.8	74.4	74.9	75.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	80.3	80.2	80.8	80.8	79.6	78.3	77.9	78.0	78.0	78.6	79.2	(27)	
		MCDB	86.2	86.4	86.9	86.1	85.7	84.3	83.7	82.7	83.7	85.5	86.2	86.4	(28)	
	2%	WB	79.2	79.2	79.3	79.8	79.6	78.3	77.2	76.7	76.8	76.9	77.6	78.3	(29)	
		MCDB	85.5	85.6	85.9	85.6	85.1	83.7	82.7	82.7	83.4	85.0	85.6	85.9	(30)	
	5%	WB	78.4	78.7	78.9	79.1	78.9	77.5	76.3	75.9	76.0	76.1	76.7	77.7	(31)	
		MCDB	84.9	85.1	85.4	85.1	84.6	82.9	81.7	81.8	82.7	83.9	84.9	85.2	(32)	
	10%	WB	77.7	78.0	78.3	78.6	78.2	76.6	75.5	75.0	75.3	75.5	76.1	77.0	(33)	
		MCDB	84.2	84.5	84.8	84.6	83.7	81.8	80.7	80.9	82.1	82.9	83.8	84.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.0	12.1	12.1	12.3	12.3	12.6	13.6	13.4	12.3	11.7	11.5	(35)	
		MCDBR	12.6	12.8	13.3	13.3	13.8	13.9	14.1	14.7	14.4	12.9	12.8	12.5	(36)	
	5% WB	MCWBR	4.3	4.3	4.6	4.9	5.2	5.5	5.7	5.5	5.0	4.2	4.1	4.1	(37)	
		MCWBR	11.4	12.0	12.3	12.3	12.5	12.4	12.6	13.2	13.3	12.4	12.1	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.401	0.406	0.399	0.407	0.391	0.392	0.393	0.398	0.420	0.401	0.403	0.412	(40)	
		taud	2.486	2.477	2.514	2.506	2.545	2.537	2.518	2.469	2.379	2.437	2.439	2.433	(41)	
	Ebn at Noon		297	295	293	283	279	273	275	281	282	293	294	292	(42)	
		Edh at Noon	37	37	35	34	31	31	32	35	40	38	38	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1911	1912	1866	1683	1595	1446	1494	1758	1926	2039	2079	1979	(44)	
(45)		RadStd	127	103	97	107	106	80	102	60	85	69	63	69	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-0.86	N/A	-0.56	-0.54	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	-0.86	N/A	-0.56	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page